TEACHER COMPANION / EDITION 1.0

Teach with the book

Suggested answers and evidence guidance for the 50-page Footstool student workbook.

Use the course site for explanation, demonstrations, presentations and teacher-led video. Students can read, discuss, sketch and record their evidence in the workbook. This guide supports feedback; it does not supply formal marks, outcomes, due dates or testing instructions.

THE CONTROLLING SOURCES

The unchanged original two-page Wagga High School Foot Stool PDF is the sole dimensional and construction authority. Workbook pages 43-44 reproduce it at a resized page scale: read the written dimensions, not the printed line lengths. Current school SOPs and teacher directions control all practical work. Slide 19 of the source deck is excluded because it conflicts with the original plan.

Find the answer key

Module	Learning focus	Student pages	Guide page
1	Workshop readiness	3-6	2
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Guide page 12 covers the original-plan and project-record pages. Page 13 preserves the full optional video inquiry. Page 14 maps the folio and records sources and illustration credits.

Open the Footstool course

Open the unchanged original two-page plan

MODULE 1 / STUDENT PAGES 3-6

Workshop readiness

Readings: pages 3-4. Check: page 5. Studio: page 6.

Workbook page 5 / multiple-choice key

Item	Key	Correct answer and teaching point
Q1	**B**	**To reduce unexpected movement and crowding** Staying in authorised areas helps control movement, crowding and unsafe choices.
Q2	**D**	**Elimination** Elimination is strongest because it removes the hazard completely.
Q3	**A**	**Understanding the instruction, checking the area and having permission** Readiness includes understanding the instruction, checking the area and having teacher permission before starting.

Page 5 / Build the control ladder

Row order: engineering; elimination; administrative; substitution; PPE.

Removing the loose object eliminates that hazard. A warning changes behaviour while leaving the object present. Strong explanations connect each control to the risk it changes.

Page 6 / starting point and readiness

Know / Wonder / Learned: Accept the student’s authentic starting knowledge and a specific question. Learned is completed later using an accurate idea or changed understanding. Do not require students to invent an initial answer.

Vocabulary confidence: Self-ratings are diagnostic, not marks. Accept relevant examples or precise questions for unfamiliar terms. Revisit the vocabulary during later modules; do not count uncertainty as an incorrect claim.

Readiness and goal: Look for one achievable learning goal and two justified conduct actions, such as keeping the route clear, staying in an authorised area, waiting for permission, reporting damage, or asking before acting. Each reason must connect the action to another person or a recognisable risk.

Reading discussion: A poorly placed item can obstruct the route or interfere with another person. Accept an observed classroom example; do not invent a local hazard.

Open Module 1: readings, checks and source activities

MODULE 2 / STUDENT PAGES 7-10

Tools and accurate references

Readings: pages 7-8. Check: page 9. Studio: page 10.

Workbook page 9 / multiple-choice key

Item	Key	Correct answer and teaching point
Q1	**C**	**Pause and ask for the approved name** Pausing and asking prevents inaccurate identification and unsuitable choices.
Q2	**A**	**The written dimensions on the original plan** Written dimensions on the original plan provide the dimensional authority for the project.
Q3	**D**	**Whether teacher authorisation and the current SOP apply** Teacher authorisation and the current SOP must apply before tool use.

Page 9 / Tools in disguise

ECVI = vice; EOTNN ASW = tenon saw; LIDRL = drill; RAQSEU = square.

Page 10 / gallery identification

Image	Approved name	Broad purpose
A	Combination square	Square measuring or marking reference.
B	Tenon saw	Cuts wood with a backed saw.
C	Battery drill	Produces holes using approved drilling equipment.
D	Bevel-edge chisel	Removes small amounts under teacher direction.
E	Woodwork vice	Holds work securely when teacher approved.

Accept a recognisable labelled tool feature, an accurate broad purpose and a precise question. Readiness also requires the correct task information, teacher permission, applicable procedure, suitable condition and clear area.

Optional source scrambles: ELSIHC = chisel; RSENDA = sander; EPLAN = plane; CSERW = screw; MEHAMR = hammer; LNAI = nail; TLEALM = mallet; LIPECN = pencil.

Open Module 2: readings, checks and source activities

MODULE 3 / STUDENT PAGES 11-14

Timber properties

Readings: pages 11-12. Check: page 13. Studio: page 14.

Workbook page 13 / multiple-choice key

Item	Key	Correct answer and teaching point
Q1	**D**	**Softwood** Pine belongs to the softwood botanical group.
Q2	**B**	**One surface may not show the complete grain pattern** Different faces can reveal different parts of the grain pattern.
Q3	**C**	**All softwood is always easy to work** A botanical category alone does not prove how workable every sample will be.

Page 13 / Timber property matcher

Row order: hardwood; straight-grained; workability; knot; softwood; dense.

Hardwood and softwood describe botanical groups, not hardness ratings. Density relates mass to volume. Workability depends on the material, its condition, the approved tool and the process.

Page 14 / sample field notes

Sample field notes: Accept accurate observations tied to the actual sample. A student may report no visible knot or insufficient density evidence. Look for grain direction/pattern, a location-specific knot or texture observation, and supported density language. Do not require invented test results.

Pine explanation: Pine is the course-identified project timber and a softwood botanical group member. Suitability requires actual material properties, condition and teacher requirements; the word softwood does not establish hardness, strength or uniform behaviour. Workability depends on the material, approved tool and process.

Reading discussion: Accept checkable descriptions such as straight or wavy grain and smooth or uneven texture, when supported by the actual sample. Coniferous means cone-bearing; it is classification language, not a performance claim.

Open Module 3: readings, checks and source activities

MODULE 4 / STUDENT PAGES 15-18

Engineered wood

Readings: pages 15-16. Check: page 17. Studio: page 18.

Workbook page 17 / multiple-choice key

Item	Key	Correct answer and teaching point
Q1	**A**	**Wood elements are combined through controlled manufacturing processes** Engineered wood products combine wood layers, particles or fibres through controlled processes.
Q2	**C**	**Repeated thin layers along the edge** Repeated veneer layers along the edge are a key structural clue for plywood.
Q3	**B**	**Condition and surface quality** Condition and surface quality are relevant material features that can affect suitability.

Page 17 / Layer detective and the material claim

Row order: plywood; particleboard; MDF. These identify broad structures; a real uncertain sample still requires an approved comparison.

A stronger environmental claim considers sourcing, manufacture, adhesives, transport, useful life, maintenance and waste. Small wood elements alone do not prove the most sustainable option.

Page 18 / board investigation

Look for sample-based descriptions of structure, face, edge and condition. Link one feature to a possible use and state the supporting information or uncertainty. A surface alone may not establish the material.

Wood forms: Local theory supports layers/veneers, particles and fine fibres. Additional forms require evidence from the teacher-shown source.

LVL and climate questions: Teacher to confirm from the actual source video or approved extract. Require attribution and cautious language. The local readings do not provide a full LVL manufacturing answer or quantified climate claim.

The complete twelve-question source-video extension is on guide page 13.

Open Module 4: readings, checks and source activities

MODULE 5 / STUDENT PAGES 19-22

Read the original plan

Readings: pages 19-20. Check: page 21. Studio: page 22.

Workbook page 21 / multiple-choice key

Item	Key	Correct answer and teaching point
Q1	**B**	**It may not show all information clearly or accurately** A thumbnail may help locate information but may not show every detail clearly or accurately.
Q2	**D**	**To provide communication cues** Symbols communicate meaning efficiently and guide the reader towards relevant information.
Q3	**A**	**The unchanged original two-page PDF** The unchanged original PDF is the authoritative reference for plan information.

Page 21 / Rebuild the reading check

Sequence: B, D, E, A, C. Row entries: A = 4; B = 1; C = 5; D = 2; E = 3.

Confirm title/page; locate the feature and view; read its linked information; compare the related view; clarify uncertainty. A student may need to pause earlier if the information is unclear.

Page 22 / original-plan reading record

Original-plan reading record: Check the title, page, view and feature against the unchanged two-page Footstool PDF. Any copied dimension or symbol must belong to the identified feature; do not accept values derived from scaling a preview. This guide deliberately supplies no project dimensions or assumed joints.

Related views and explanation: Look for consistent identification of one feature across views and an explanation of what each view contributes. The student should connect written information to its feature and state a precise uncertainty or clarification. Original PDF and teacher direction remain controlling.

Reading discussion: Scale is a proportional representation. Written dimensions control intended size. Symbols, notes and dimensions must be interpreted with the correct feature and view.

Open Module 5: readings, checks and source activities

MODULE 6 / STUDENT PAGES 23-26

Graphical communication

Readings: pages 23-24. Check: page 25. Studio: page 26.

Workbook page 25 / multiple-choice key

Item	Key	Correct answer and teaching point
Q1	**D**	**To identify lines representing the same part** Alignment helps connect representations of the same feature across related views.
Q2	**C**	**They are guides and should not look like object outlines** Light lines guide drawing and alignment without being confused with visible object edges.
Q3	**B**	**It combines several visible faces in one view** An isometric drawing combines several visible faces instead of separating them into individual views.

Page 25 / line classification and photocopy question

Row order: H - hidden detail; C - centre line; D - dimension line; V - visible outline.

The written dimensions on the unchanged original plan still control intended size. A larger photocopy changes apparent line lengths, not those values.

Page 26 / drawing communication evidence

Look for a labelled isometric communication sketch with consistent principal directions: vertical and two receding directions commonly about 30 degrees to the horizontal. Matching edge directions remain parallel.

Third-angle views separate directions so features can be compared; an isometric view combines several visible faces in one image. The student should trace one identified feature to a related view and explain what the comparison confirms or leaves uncertain.

Accept a precise unanswered question when necessary. Judge the clarity of communication and source checking, not invented dimensions or an assumed construction method.

Open Module 6: readings, checks and source activities

MODULE 7 / STUDENT PAGES 27-30

Personal emblem design

Readings: pages 27-28. Check: page 29. Studio: page 30.

Workbook page 29 / multiple-choice key

Item	Key	Correct answer and teaching point
Q1	**A**	**Combining and adapting useful reference features** Combining and adapting reference features shows independent design thinking.
Q2	**B**	**They explore different main ideas or arrangements** Distinct concepts explore meaningfully different symbols, combinations or arrangements.
Q3	**C**	**Simplifying a distracting outline to improve clarity** Simplifying a distracting outline is a purposeful improvement linked to clarity.

Page 29 / Name the criterion

Row order: recognition; balance; contrast; clarity.

For “I chose it because it looks good”, require one observable strength linked to a criterion and one limitation. A hypothetical example is: “The main symbol is easy to recognise, but the surrounding detail weakens clarity.” The student must use their own actual evidence when evaluating their work.

Page 30 / emblem studio evidence

Expect three meaningful themes with reference sources and useful observations; three distinct concepts with arrangement, meaning, strength and limitation annotations; a justified selection; and a labelled refinement linked to a criterion.

Different concepts change the main idea, combination or arrangement, rather than merely changing size. Refinement may simplify an outline, adjust balance, strengthen contrast or remove distracting detail while keeping meaning.

> **TEACHER CONFIRMATION**
>
> Laser cutting and engraving are optional. A concept or refinement does not authorise a decorative process or provide settings.

Open Module 7: readings, checks and source activities

MODULE 8 / STUDENT PAGES 31-34

Progress and quality evidence

Readings: pages 31-32. Check: page 33. Studio: page 34.

Workbook page 33 / multiple-choice key

Item	Key	Correct answer and teaching point
Q1	**D**	**A visible gap remained during the check** This statement reports a visible condition without adding an unsupported judgement.
Q2	**A**	**Specific information that can be observed or checked** Quality evidence is specific and can be supported through observation, checking, feedback or an approved photograph.
Q3	**B**	**The recorded observation** The action should address the specific observation recorded in the reflection.

Page 33 / reflection sequence

Row numbers: 3, 1, 4, 2. The sequence is observation, possible reason, action and next teacher-confirmed step. Underline “may” and “not yet confirmed”.

The student still needs teacher-confirmed direction before a practical correction. In the fictional log, confirmation of further checking does not authorise a correction method.

Page 32 / replace “Everything is finished”

Accept a precise question that identifies a feature or requirement and asks what evidence confirms completion. Example: “Which check of this feature is still needed before you can confirm this stage is complete?” Do not require or invent a completed result.

Page 34 / quality case file

Expect lesson context, a broad task, the feature and reference checked, a factual observation, an approved photo or labelled sketch, and a caption explaining both visible evidence and its limits.

A possible cause stays cautious until confirmed. Distinguish an action already taken from a proposal; record the confirmed next step or the exact question awaiting an answer. Evidence may remain pending if the work has not reached an appropriate observation.

Open Module 8: readings, checks and source activities

MODULE 9 / STUDENT PAGES 35-38

Function and testing

Readings: pages 35-36. Check: page 37. Studio: page 38.

Workbook page 37 / multiple-choice key

Item	Key	Correct answer and teaching point
Q1	D	**Designed to support suitable and comfortable human use** Ergonomic design considers how a product supports suitable and comfortable human use.
Q2	C	**Evidence about the criterion it was designed to examine** One check may support a limited judgement about its chosen criterion without proving all other qualities.
Q3	A	**Using cautious language linked to a functional criterion** A prediction should be cautious and explain which functional quality may be affected.

Page 37 / vocabulary bridge

Term	Similar meaning	Opposite
Durable	long-lasting / able to withstand expected use	short-lived / easily worn out
Reliable	consistent / dependable	inconsistent / unreliable
Stable	steady / resists unwanted movement	unsteady / wobbly
Ergonomic	suitable and comfortable for human use	awkward or uncomfortable for the intended user
Efficient	uses time, effort and resources without unnecessary waste	wasteful / inefficient
Economical	good value considering cost and resource use	poor value / uneconomical

Page 38 / functional investigation

Expect one explained criterion, an identified teacher-approved check, a specific observation, a limited judgement, an evidenced limitation and a connected improvement proposal. Predictions remain cautious and the next action needs confirmation.

The human-use lens requires suitable, comfortable-use evidence or an honest statement that it is absent. Efficient is not simply fastest; economical is not simply cheapest. Do not invent a load, user, force, test method, pass level or plan change.

Open Module 9: readings, checks and source activities

MODULE 10 / STUDENT PAGES 39-42

Aesthetics and final evaluation

Readings: pages 39-40. Check: page 41. Studio: page 42.

Workbook page 41 / multiple-choice key

Item	Key	Correct answer and teaching point
Q1	D	**A natural, irregular or flowing shape** Organic forms are inspired by natural, irregular or flowing shapes.
Q2	B	**Visual hierarchy** Visual hierarchy describes what a viewer notices first, second and later.
Q3	C	**I learned to separate observations from opinions and can use clearer evidence next time** This reflection identifies specific learning and explains how it can improve future work.

Page 41 / twelve design clues

Left clue	Answer	Right clue	Answer
Something that works consistently	**reliable**	Opposite to safe	**dangerous**
Opposite to wasteful	**efficient**	A repeated image or arrangement	**pattern**
A strong difference, such as black and white	**contrast**	A mirrored or evenly balanced image	**symmetrical**
Opposite to wobbly	**stable**	Makes something more noticeable or helps it stand out	**emphasis**
Lasts a long time	**durable**	Comfortable and suitable for the hand or body	**ergonomic**
How something works	**function**	How something looks	**aesthetics**

Page 42 / balanced final evaluation

Look for a supported functional strength; two visual qualities explained through observable features and effects; an evidenced limitation; a realistic improvement and its predicted effect; and specific learning that can transfer to future work.

All twelve self-ratings are optional reflection prompts, not marks or pass claims. Leave unexamined qualities unrated. Accept different aesthetic responses when the feature, criterion and explanation are coherent. Symmetry is more precise than general visual balance.

Open Module 10: readings, checks and source activities

STUDENT PAGES 43-50

Plans and project records

Use actual student evidence. A blank or uncertain result is preferable to an invented observation or approval.

Pages 43-44 / original plan: Check each copied dimension against the exact feature and view on the original PDF. For the view comparison, require one feature identifiable in both views and an explanation of what each contributes. No numerical answer is preselected in this guide.

Page 45 / approved record: Materials, component information, class instructions and the next practical task must match actual teacher-issued information. Teacher initials and date are completed by the teacher; the page itself creates no approval.

Pages 46-47 / journal and image: Look for dated broad tasks, a requirement checked, observations, an action with a reason and a next step or question. Image labels A and B must identify visible features and explain what the image cannot prove.

Page 48 / purposeful viewing: Require an identified video, an enquiry question, traceable observations or claims and an explained course connection. Separate demonstrated evidence from unsupported claims and identify any limitation.

Page 49 / eight-word lexicon

Clue	Answer
Direction and visible pattern of wood fibres	**grain**
Evidence of where a branch grew	**knot**
Describes a cone-bearing tree	**coniferous**
Suited to the user and their body	**ergonomic**
Ability to keep performing over time	**durability**
Achieves a purpose with little waste	**efficient**
Pictorial view showing three directions of a form	**isometric**
Visual qualities and appearance	**aesthetics**

Accept a paragraph using three terms accurately in the student's own context. Classification and appearance words must not become unsupported performance claims.

Page 50 / final synthesis: Require one evidenced functional judgement, two visible aesthetic qualities, a justified improvement with checks still needed, and an explained skill development. Do not invent a finish, result or submission requirement.

OPTIONAL TEACHER-LED EXTENSION

Extend the video inquiry

The source deck's twelve questions are preserved below. Student page 18 selects three; page 48 supports additional evidence notes.

Source video: Manufactured Wood Products

01. Define manufactured wood products. What other name is commonly used for these products?

02. What types or forms of wood are used in manufactured wood products?

03. Which two wood properties are considered important when developing new manufactured wood products?

04. Explain how plywood is made.

05. What property makes plywood and oriented strand board suitable for structural wood panelling?

06. How is laminated veneer lumber made?

07. What makes laminated veneer lumber useful compared with many solid wood products? Give examples of uses.

08. How is cross-laminated timber made? Give examples of uses.

09. Which manufactured wood product can substitute for load-bearing timber across wide spans in multi-storey buildings?

10. How is wood-plastic composite made? Give examples of uses.

11. How can using and substituting with wood products help address climate change?

12. Which manufactured wood product was used as the main structural material in the Forté building shown in the video?

What can be answered from the local reading?

Questions 1, 2 and 4: Manufactured or engineered wood combines wood elements through controlled processes. Local examples use layers/veneers, particles and fine fibres. Plywood uses bonded veneer layers, usually with changing grain directions between neighbours.

> **TEACHER TO CONFIRM FROM THE ACTUAL SOURCE**
>
> Confirm video-specific answers and examples for Questions 3 and 5-12 from the teacher-shown video or approved extract. This includes the two named properties, structural claims, LVL/CLT/WPC manufacture and uses, climate claims and Forté identification. Require evidence and attribution; do not infer these answers from a product photograph.

SOURCE AND FOLIO NOTES

Keep the evidence connected

The workbook follows the current ten-module website and supports the existing twelve-card folio.

Folio card	Evidence focus	Student pages
01	Starting point	6
02	Workshop readiness	3-6
03	Tool communication	7-10
04	Timber language	11-14
05	Board investigation	15-18, 48
06	Plan reading	19-22, 43-44
07	Drawing communication	23-26
08	Emblem concepts	27-30
09	Design choice	28-30
10	Quality log	31-34, 45-47
11	Function review	35-38, 50
12	Aesthetic review	39-42, 50

Sources and illustration credits

Theory and the 30 selected multiple-choice items come from the website's 30 named section files. Activity prompts follow its source-activities and folio resources and the original 25-slide activity deck. Slide 19 supplies no workbook technical content.

Three generated photographs provide illustrative project, timber-grain and emblem context; they are not construction drawings or evidence of a student's result. Original vector diagrams explain general concepts. The five source tool photographs retain the course image-library provenance documented in VISUAL-AUDIT.md: copies from the Construction-Supplementary-material repository's Construction Pictures and additional tools folders. No new licence claim is made.

The original Wagga High School plan pages are reproduced without content changes and resized to fit. Their written dimensions remain authoritative. Student evidence, approval records and self-ratings are completed from the actual classroom situation.

Open the source folio

Open teacher resources and the source plan/activity deck